

50 MCQs and 3 Code Tasks

MCQs:

1. Which register is automatically decremented when the LOOP instruction is executed?
2. Which instruction is used to bypass a loop when CX contains zero at the start?
3. Flow control instructions are primarily used to manage which of the following?
4. Which type of loop guarantees at least one execution of its body?
5. The instruction JCXZ label transfers control when CX is equal to which value?
6. The LOOP instruction transfers control to the target label until which condition becomes true?
7. In 8086 assembly, the INT 21H service with AH = 1 performs what operation?
8. In an AND condition, the entire condition is true when:
9. In an OR condition, the condition becomes false only when:
10. Which loop checks its termination condition first?
11. Which loop checks its termination condition at the end?
12. In FOR loop implementation, the counter variable is stored in which register?
13. What happens if CX = 0 before entering a loop without the JCXZ instruction?
14. The label after a jump or loop typically marks what?
15. In assembly, conditional jumps are based on the values in which register(s)?
16. The instruction CMP AL, 'A' affects which flag(s)?
17. Which of the following instructions is used to display a character in DOS interrupt 21h?
18. In an AND condition, which instruction would typically appear first?
19. The pseudo-code IF condition corresponds to which assembly instruction?
20. In the example of checking uppercase letters, which ASCII range is tested?
21. The instruction JG label jumps when which condition occurs?
22. Which interrupt is used for keyboard input in the slide examples?
23. The WHILE loop in assembly usually starts with which type of instruction?
24. In a REPEAT loop, the condition is checked using which instruction?

25. A WHILE loop contains how many jumps in typical assembly translation?
26. A REPEAT loop contains how many conditional jumps?
27. The instruction INC DX performs what operation?
28. Which flag is used by the JE instruction?
29. In the REPEAT example, which character terminates the loop?
30. The carriage return character has which hexadecimal ASCII value?
31. In FOR loops, which instruction is responsible for automatic decrementing?
32. The LOOP instruction will jump only when which condition is met?
33. To skip loop execution entirely when CX=0, which instruction is used?
34. The JCXZ instruction checks which register?
35. Which type of loop is considered count-controlled?
36. Which loop structure is shortest in assembly?
37. The pseudo-code UNTIL condition corresponds to which assembly style?
38. Which register is used to store the character read from the keyboard?
39. The conditional jump JL label triggers when:
40. To display a star (*) repeatedly, which register must contain the ASCII value of '*'?
41. OR conditions are implemented using which type of jumps?
42. AND conditions typically require how many conditional checks?
43. For MOV AH,2, the character to display must be stored in which register?
44. Which loop type is preferred when the number of iterations is known beforehand?
45. In WHILE loops, initialization must be done:
46. The unconditional jump instruction is:
47. The instruction used to compare two values in assembly is:
48. Which loop runs at least once regardless of the initial condition?
49. In the uppercase detection example, what happens if the character is not between A and Z?
50. The label END_IF: in the slide marks the point where:

Code Tasks:

1. Write an 8086 assembly program that reads a character and prints it only if it is '1' or '2'; otherwise terminate.
2. Write an 8086 assembly program using a WHILE loop to count vowels until a period '.' is typed.
3. Write a FOR-loop based 8086 program that takes input N and prints '#' N times.